Department of Information Systems and Technologies

CTIS 365: Applied Data Analysis

Semester: SPRING 2025-2026

Lab Guide #4

OBJECTIVES: Central Tendency, Variability
Instructor: Asst. Prof. Seyid Amjad ALI **Assistant:** Efe Mert SAHINKOC

Q1. Use the given vector that keeps the ages.

Find the central tendencies and variability. Decide which measurement for central tendency and variability are more appropriate.

```
vector  <- c(26, 23, 24, 29, 31, 28, 32, 21, 32, 29, 25, 33, 27, 34)
```

Q2. Use **skatingTime.txt**

a. Find the central tendencies of skating times.

b. Sketch a histogram for skating times.

Q3. A **population** dataset named **"q3data.txt"** is given to you

a. Find the mean of column V1.

b. Find variance and standard deviation by using 2 different approaches and verify that they all lead to the same result (computational formula and R's built-in function) for sample data.

c. Add 2 points to each score and compute the new mean, variance and standard deviation to see the changes.

Q4. Use the **happiness.csv**

a. Construct a bar graph showing the 3 big cities (Ankara, Istanbul, Izmir) happiness level according to the genders.

b. Find the central tendencies for females and males happiness Level.